

Unit 4 Notes

Properties of Polygons and Quadrilaterals

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Unit 4 Notes: Properties of Polygons and Quadrilaterals

Big goal

Polygons and quadrilaterals are classified by side, angle, diagonal, and parallel-line properties. Coordinate geometry lets you prove these properties instead of relying on a picture.

1. Polygon angle facts

An n -gon is a polygon with n sides. A diagonal connects two nonconsecutive vertices. Drawing diagonals from one vertex divides an n -gon into $n - 2$ triangles.

Polygon formulas

Interior angle sum of an n -gon:

$$(n - 2)180^\circ.$$

Each interior angle of a regular n -gon:

$$\frac{(n - 2)180^\circ}{n}.$$

Exterior angle sum of any convex polygon:

$$360^\circ.$$

Each exterior angle of a regular n -gon:

$$\frac{360^\circ}{n}.$$

2. Quadrilateral family

A quadrilateral has four sides. Important special quadrilaterals include parallelograms, rectangles, rhombi, squares, trapezoids, isosceles trapezoids, and kites.

Figure	Defining properties
Parallelogram	Both pairs of opposite sides are parallel.
Rectangle	A parallelogram with four right angles. Diagonals are congruent.
Rhombus	A parallelogram with four congruent sides. Diagonals are perpendicular and bisect opposite angles.
Square	A rectangle and a rhombus: four congruent sides and four right angles.
Trapezoid	Exactly one pair of opposite sides is parallel, using the common high-school definition.
Isosceles trapezoid	A trapezoid with congruent legs. Base angles and diagonals are congruent.
Kite	Two distinct pairs of adjacent congruent sides. One diagonal is a perpendicular bisector of the other.

3. Parallelogram theorems

If a quadrilateral is a parallelogram, then:

- opposite sides are congruent;
- opposite angles are congruent;

- consecutive angles are supplementary;
- diagonals bisect each other.

Converses are also useful. A quadrilateral is a parallelogram if either both pairs of opposite sides are parallel, both pairs of opposite sides are congruent, one pair of opposite sides is both parallel and congruent, or the diagonals bisect each other.

4. Special parallelograms

A rectangle is a parallelogram with congruent diagonals. A rhombus is a parallelogram with perpendicular diagonals or with all sides congruent. A square satisfies both rectangle and rhombus properties.

Most specific name

Always give the most specific classification supported by the evidence. A square is also a rectangle, rhombus, parallelogram, and quadrilateral, but its most specific name is square.

5. Coordinate classification

Coordinate geometry provides evidence:

- Use slopes to test parallel and perpendicular sides.
- Use distances to test congruent sides and diagonals.
- Use midpoints to test whether diagonals bisect each other.

Coordinate classification plan

1. Compute slopes of all sides. Look for parallel pairs and right angles.
2. Compute side lengths. Look for congruent sides.
3. Compute diagonal lengths and diagonal midpoints.
4. Use the strongest conclusion: square, rectangle, rhombus, parallelogram, trapezoid, kite, or general quadrilateral.

6. Perimeter and area on the coordinate plane

Perimeter is the sum of side lengths, so use the distance formula. Area can often be found by splitting a figure into rectangles and triangles. For coordinate polygons, the shoelace formula is efficient:

$$A = \frac{1}{2} |(x_1y_2 + x_2y_3 + \cdots + x_ny_1) - (y_1x_2 + y_2x_3 + \cdots + y_nx_1)|.$$

For triangles and rectangles, base-height reasoning is often faster.

7. Congruent triangles inside quadrilaterals

Many quadrilateral properties come from triangle congruence. A diagonal of a parallelogram creates two congruent triangles. CPCTC then proves opposite sides and opposite angles congruent.

8. Worked examples

Example 1: Polygon angles

For a regular decagon, each exterior angle is $360^\circ/10 = 36^\circ$. Each interior angle is $180^\circ - 36^\circ = 144^\circ$.

Example 2: Parallelogram algebra

In a parallelogram, consecutive angles are supplementary. If they measure $3x + 10$ and $5x - 6$, then

$$3x + 10 + 5x - 6 = 180 \Rightarrow 8x + 4 = 180 \Rightarrow x = 22.$$

The angles are 76° and 104° .

Example 3: Coordinate rectangle

For $A(0, 0)$, $B(6, 0)$, $C(6, 4)$, $D(0, 4)$, opposite sides have slopes 0 and undefined in matching pairs, and adjacent sides are perpendicular. The figure is a rectangle. Since side lengths are 6 and 4, it is not a square.

Mastery checklist

You are ready for Unit 4 when you can use polygon angle formulas, state parallelogram and special quadrilateral properties, prove quadrilaterals from coordinates, compute perimeter, and find area by decomposition or coordinate methods.